knowledge thus far accumulated by physical science as being of a lasting character.

What then is the substance of these findings? First of all, it must be noted that all data of physical knowledge are founded on measurements, and that every measurement takes place in space and time with the orders of magnitudes varying from the inconceivably vast to the infinitesimally small. We can get an approximate idea of the distance of the cosmic regions from which a message can still reach us if we consider that light, which traverses the distance from the Moon to the Earth in a second or so, requires many millions of years to arrive from those regions to our planet. On the other side of the picture, physical science must calculate with spatial and temporal magnitudes, the infinite smallness of which can be realized by the comparison of a head of a pin with our entire planet.

Measurements of the most varied kinds have been consistent in leading to the conclusions that all physical occurrences without exception can be reduced to mechanical or electrical processes, produced by the movements of certain elementary particles, such as electrons, positrons, protons and neutrons; and both the mass and the charge of all of these particles are each expressed by an extremely small but quite definite number. The precision of this number increases with improvements in the accuracy of methods of measurement. These minute numbers, the so-called universal constants, are in a sense the immutable building blocks of the edifice of theoretic physics.

So now we must continue with the question: What is the real meaning of these constants? Are they, in the last analysis, inventions of the inquiring mind of man, or do they possess a real meaning independent of human intelligence?

The first of these two views is professed by the followers of Positivism, or at least by its most extreme partisans. Their theory is that physical science has no other foundation than the measurements on which its structure is erected, and that a proposition in physics makes any sense only insofar as it can be supported by measurements. But since every measurement presupposes an observer, from the positivistic viewpoint the real substance of a law of physics can never be detached from the observer, and it loses its meaning as soon as one attempts mentally to eliminate the observer and to see something more, something real, behind him and his measurement.

This outlook cannot be challenged from the purely logical viewpoint. And yet, a closer examination must brand this version of it as inadequate and unproductive. For it disregards a circumstance which is of a decisive importance in the extension and progress of scientific knowledge. However much Positivism may regard itself as proceeding without presuppositions, it is nonetheless committed to a fundamental premise if it is not to degenerate into an unintelligible solipsism. This premise is that every physical measurement